# DSCI 554 - Practice Quiz 1

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# Quiz

## Question 1

**In one or two sentences and your own words**, explain how experimental studies differ from observational studies.

## Question 2

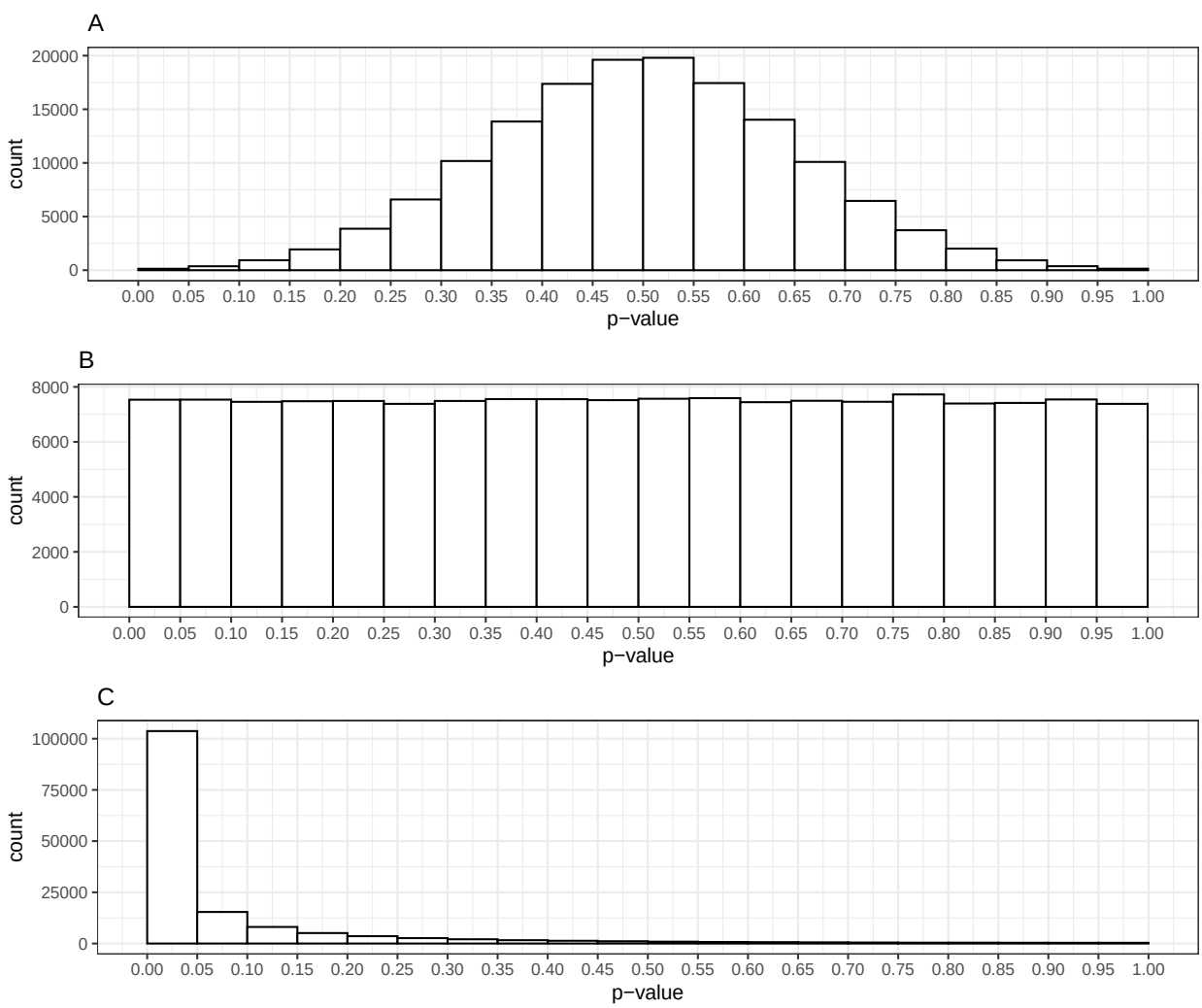
Consider the following two hypotheses:

$$H_0 : \text{Tweets containing a } \texttt{gif} \text{ have the same average number of impressions as tweets that do not contain a } \texttt{gif}.$$

$$H_a : \text{Tweets containing a } \texttt{gif} \text{ have a different average number of impressions than tweets that do not contain a } \texttt{gif}.$$

We ran two simulations, one where $H_0$ was true and one where $H_0$ was not true. In each simulation, we ran 150,000 $t$-tests on **different datasets** and created a histogram to examine the distribution of the $p$-values of each simulation.

**Match the correct distribution** of the p-values to the simulation scenario where $H_0$ was true and to the simulation scenario where $H_0$ was not true.



*The histogram labelled letter [A, B, **OR** C] matches the simulation scenario where $H_0$ was true.*

*The histogram labelled letter [A, B, **OR** C] matches the simulation scenario where $H_0$ was not true.*

## Question 3

**To answer this question, take Question 2 as the context.**

Using your own words, re-write the inferential statement in $H_a$ to make it causal.

## Question 4

What is a confounder? Why is it important to consider such variables in your analysis? **Answer in two to three sentences and in your own words.**

## Question 5

How does randomization contribute to establishing a causal relationship in the context of an experimental study? **Answer in two to three sentences and in your own words.**

## Question 6

Is the following an example of an experimental or observational study?

> *A gym wants to collect evidence to show that their new Zumba class reduces body fat percentage compared to the original Zumba class. They randomly select 30 gym-goers who just signed up for this new Zumba class, as well as 30 gym-goers who just signed up for their original Zumba class. Then, they measure the body fat percentage of all randomly selected gym-goers before and after ten weeks of attending these classes.*

## Question 7

Dataset `ToothGrowth` is an experiment related to the effect of Vitamin C on tooth growth in guinea pigs. The response is the length of odontoblasts (in picometers, pm). Each guinea pig was randomly allocated to one of three dose levels of vitamin C (`0.5`, `1`, and `2` mg/day) by one of the two delivery methods: orange juice (`OJ`) and ascorbic acid (a form of Vitamin C coded as `VC`).

```r
# Copy and paste this chunk of code in your R console.
data(ToothGrowth)
ToothGrowth <- ToothGrowth %>%
  mutate(dose = as.factor(dose))
head(ToothGrowth)
```

```
##    len supp dose
## 1  4.2   VC  0.5
## 2 11.5   VC  0.5
## 3  7.3   VC  0.5
## 4  5.8   VC  0.5
## 5  6.4   VC  0.5
## 6 10.0   VC  0.5
```

```r
levels(ToothGrowth$dose)
```

```
## [1] "0.5" "1"   "2"
```

```r
levels(ToothGrowth$supp)
```

```
## [1] "OJ" "VC"
```

The main inquiry in this experiment is the following:

> *Is there any statistically significant combination of dose level and delivery method that maximizes the length of odontoblasts compared to the baseline dose of `0.5` mg/day and orange juice? If so, by how much?*

Based on the previous framework, answer the following:

1. How many treatments do we have in this experiment? **Answer briefly.**
2. In an experimental framework, what is the dose combination of `0.5` mg/day and orange juice? **Answer briefly.**
3. Is this a balanced or unbalanced experiment? Why or why not? **Answer in one or two sentences.**

## Question 8

**To answer this question, take Question 7 as the context.**

How would you statistically analyze this experimental data to answer the main inquiry? **Answer in one or two sentences** while justifying your model.

Then, **fit the model** and **provide the corresponding code**.

## Question 9

**To answer this question, take Questions 7 and 8 as the context.**

Provide the corresponding code and output to globally assess the statistical significance of the modelling component associated with the main inquiry. Then, make your corresponding inferential conclusion with $\alpha = 0.05$.

Based on the previous conclusion, if necessary with $\alpha = 0.05$, execute the corresponding post-hoc tests to address this main inquiry fully (provide the corresponding code and output). Make sure to use a conservative correction method.

# Solution Key

## Question 1

*Experimental studies differ from observational studies in that the researcher manipulates/assigns the treatment to the experimental units and then observes the effects of the treatments on the response variable (treatments should be randomly assigned). On the other hand, observational studies collect the corresponding sampled data (response and explanatory variables) from the population of interest without manipulating any independent variable.*

## Question 2

*The histogram labelled letter* ***B*** *matches the simulation scenario where* $H_0$ *was true.*

*The histogram labelled letter* ***C*** *matches the simulation scenario where* $H_0$ *was not true.*

## Question 3

*Including a* `gif` *causes a tweet to receive a different average number of impressions compared to tweets that do not.*

## Question 4

*A confounder is a variable associated with both the explanatory and response variable. In an observational study, it is important to consider such variables in our analysis. Without accounting for them, we may unknowingly observe a false demonstration of an association or the masking of an actual association between the explanatory and response variable.*

## Question 5

*Randomization breaks the dependency between the factor/explanatory variable of interest and confounders. Breaking this dependency allows us to be no longer worried about confounders. Thus, we can interpret the association between the factor/explanatory variable of interest and the response variable as causal.*

## Question 6

*It is an observational study. Note that gym-goers were not randomly allocated to each type of Zumba class.*

## Question 7

1. *We have six treatments (including the control); they are all possible pairwise combinations between the levels of both factors.*
2. *It is the control treatment.*
3. *This is a balanced experiment since we have ten replicates per treatment.*

## Question 8

*We have a continuous response with two factors; thus, we can fit a two-way ANOVA model with an interaction. Including an interaction will address our main inquiry.*

```
two_way_ANOVA <- aov(formula = len ~ supp * dose, data = ToothGrowth)
```

## Question 9

*We need to obtain an ANOVA table as follows to assess the statistical significance of the double interaction globally:*

```r
library(broom)
tidy(anova(two_way_ANOVA))
```

```
## # A tibble: 4 x 6
##   term         df sumsq meansq statistic   p.value
##   <chr>     <int> <dbl>  <dbl>     <dbl>     <dbl>
## 1 supp          1  205.  205.      15.6   2.31e- 4
## 2 dose          2 2426. 1213.      92.0   4.05e-18
## 3 supp:dose     2  108.   54.2      4.11  2.19e- 2
## 4 Residuals    54  712.   13.2     NA    NA
```

*Based on the above output, we have enough statistical evidence to state that at least one pair of treatments is statistically significant for the response.*

*Now, we will use Tukey's honestly significant differences (HSD) as post-hoc tests:*

```r
tidy_tukey <- tidy(TukeyHSD(two_way_ANOVA, conf.level = 0.95)) %>%
  mutate_if(is.numeric, round, 2)
tidy_tukey
```

```
## # A tibble: 19 x 7
##    term      contrast      null.value estimate conf.low conf.high adj.p.value
##    <chr>     <chr>              <dbl>    <dbl>    <dbl>     <dbl>       <dbl>
##  1 supp      VC-OJ                  0    -3.7     -5.58     -1.82        0
##  2 dose      1-0.5                  0     9.13     6.36     11.9         0
##  3 dose      2-0.5                  0    15.5     12.7      18.3         0
##  4 dose      2-1                    0     6.37     3.6       9.13        0
##  5 supp:dose VC:0.5-OJ:0.5          0    -5.25   -10.0      -0.45        0.02
##  6 supp:dose OJ:1-OJ:0.5            0     9.47     4.67     14.3         0
##  7 supp:dose VC:1-OJ:0.5            0     3.54    -1.26      8.34        0.26
##  8 supp:dose OJ:2-OJ:0.5            0    12.8      8.03     17.6         0
##  9 supp:dose VC:2-OJ:0.5            0    12.9      8.11     17.7         0
## 10 supp:dose OJ:1-VC:0.5            0    14.7      9.92     19.5         0
## 11 supp:dose VC:1-VC:0.5            0     8.79     3.99     13.6         0
## 12 supp:dose OJ:2-VC:0.5            0    18.1     13.3      22.9         0
## 13 supp:dose VC:2-VC:0.5            0    18.2     13.4      23.0         0
## 14 supp:dose VC:1-OJ:1              0    -5.93   -10.7      -1.13        0.01
## 15 supp:dose OJ:2-OJ:1              0     3.36    -1.44      8.16        0.32
## 16 supp:dose VC:2-OJ:1              0     3.44    -1.36      8.24        0.29
## 17 supp:dose OJ:2-VC:1              0     9.29     4.49     14.1         0
## 18 supp:dose VC:2-VC:1              0     9.37     4.57     14.2         0
## 19 supp:dose VC:2-OJ:2              0     0.08    -4.72      4.88        1
```

*From the above output, using `OJ:0.5` as the control treatment (baseline dose of `0.5` mg/day and orange juice), we can see that either `OJ:2` (dose of `2` mg/day and orange juice) or `VC:2` (dose of `2` mg/day and ascorbic acid) maximize the response with estimated response differences of* $+12.8$ *and* $+12.9$ *pm, respectively. Note that both 95% confidence intervals are practically overlapped.*